

Analyzing Hallucination Patterns in RAG-Based Financial Question Answering

CS6120 NLP Project Proposal | Professor - Amir Silvio | Summer 2026 | Northeastern University

Workshop: Towards Knowledgeable Foundation Models

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Task, Goal, and Research Questions

This project investigates how and when Retrieval-Augmented Generation (RAG) systems — which ground LLM responses by first retrieving relevant passages from real documents — produce hallucinated outputs in financial question answering (QA). Even when the retrieved context is accurate, LLMs still generate answers that contradict or go beyond the source documents, a failure mode with serious consequences in finance, where a single wrong revenue figure or misattributed executive name can mislead investors and analysts. Our three research questions are: **(1)** How frequently do different RAG pipeline configurations hallucinate on financial QA? **(2)** What hallucination types dominate — numerical errors, entity misattribution, or unsupported reasoning? **(3)** Do larger or instruction-tuned models hallucinate less when given the same retrieved context?

Data

Our primary dataset is **FinanceBench** (Islam et al., 2023), a publicly available benchmark of 150 annotated QA examples drawn from real SEC filings (10-K, 10-Q, 8-K documents). Each example contains a natural-language question, a human-verified gold answer, and a pointer to the exact evidence passage — making it ideal for evaluating both answer correctness and retrieval grounding. It is freely available on HuggingFace with no collection effort required. As a secondary resource, we use **RAGBench** (Friel et al., 2024), a large multi-domain RAG evaluation benchmark with a finance subset, enabling us to examine whether hallucination patterns in finance differ from other domains. We will not create any new datasets.

Tools, Models, and Evaluation

We implement all three RAG pipelines ourselves using **LangChain**, with no off-the-shelf end-to-end RAG systems used as black boxes. The pipelines are: **(1) Baseline RAG** — BM25 sparse retrieval (rank_bm25) with fixed 512-token chunks; **(2) Dense RAG** — FAISS semantic retrieval using sentence-transformers/all-MiniLM-L6-v2 embeddings (from HuggingFace); **(3) Enhanced RAG** — dense retrieval plus LLM-based query rewriting, implemented from scratch via a prompted API call. Document preprocessing uses LlamaIndex utilities. Key hyperparameters tuned are chunk size (256 vs. 512 tokens), number of retrieved passages (top-3 vs. top-5), and generator temperature (fixed at 0.0 for determinism). For the generator, we compare **GPT-3.5-turbo** (OpenAI API) against **Mistral-7B-Instruct** (HuggingFace). Additionally, we fine-tune a **RoBERTa-base** classifier on the RAGTruth corpus to automatically categorize hallucination types — the only model we train ourselves. Evaluation uses the **RAGAS framework** (faithfulness, answer relevancy, context precision) alongside standard exact match and F1 against FinanceBench gold answers. We also manually annotate 50 failure cases and compute inter-annotator agreement (Cohen's kappa) to validate our hallucination taxonomy.

Related Work

We plan 8 papers in the proposal, expanding to 20–24 by the preliminary report. **Lewis et al. (2020)** (NeurIPS) introduces RAG and defines our architectural baseline. **Islam et al. (2023)** introduces FinanceBench, our primary benchmark. **Niu et al. (2024)** (ACL) provides RAGTruth, the hallucination corpus we use for RoBERTa supervision. **Song et al. (2024)** (EMNLP) presents RAG-HAT, the closest prior work on hallucination-aware RAG tuning, against which we compare our findings. **Friel et al. (2024)** introduces RAGBench, our secondary dataset. **Es et al. (2024)** (EACL) presents RAGAS, whose metrics we adopt directly. **Chen et al. (2021)** (EMNLP) establishes FinQA and contextualizes the difficulty of numerical reasoning in financial NLP. **Ji et al. (2025)** (NeurIPS) presents PHANTOM, the most recent work on hallucination detection in financial long-context QA, confirming the problem remains open and justifying our contribution.

Expected Results, Timeline, and Team Roles

We will produce: a main results table comparing all six pipeline/model combinations on RAGAS and F1 metrics; a bar chart of hallucination type frequencies; a qualitative case-study table with 2–3 examples per hallucination type; and a RoBERTa classification report. The most novel contribution is the manual error taxonomy — categorizing failure cases into **numerical** (wrong figures/units/years), **entity** (wrong company/executive/product), **reasoning** (logically flawed conclusions from correct premises), and **unsupported extrapolation** (claims absent from any retrieved passage) hallucinations. The 9-week timeline allocates Week 1 to proposal submission, Week 2 to baseline pipeline setup, Week 3 to dense and enhanced pipelines, Week 4 to preliminary paper (4 pages), Weeks 5–6 to Mistral experiments and manual annotation, Weeks 7–8 to full analysis and final paper, and Week 9 to poster and video. Roles are divided as: **Person 1** (Literature Lead — related work and introduction), **Person 2** (Pipeline Engineer — LangChain implementation and methodology), **Person 3** (Experiments Lead — evaluation runs, tables, figures), and **Person 4** (Analysis Lead — manual annotation, taxonomy, RoBERTa classifier, discussion). All members contribute to the final paper, poster, and video.